



Dhirubhai Ambani Institute of Information and
Communication Technology (DA-IICT)

Discrete Mathematics SC612

END SEMESTER EXAMINATION

Max Marks - 100

Autumn 2023-24

Time - 3 hours

- All questions are compulsory.
- Answers must have proper justification to get full credit.
- Understanding the questions is part of the evaluation. Don't ask for any clarification. In case of ambiguity, proceed with an assumption by clearly stating it.

- Using the Boolean identities
 - Prove that $x(x + y) = x$. [5]
 - Find the disjunctive normal form of $(x + y)\bar{z}$. [5]
- Organize the following scores in to a grouped frequency distribution table and then find the mean and standard deviation. [12]
70, 82, 56, 68, 90, 75, 58, 88, 93, 85, 65, 72, 53, 81, 84, 87, 91, 71, 67, 58, 82, 86, 72, 83, 94, 69, 84, 74, 61, 82, 83, 91, 64, 87, 84, 70, 68, 76, 89, 87, 75, 93, 80, 73, 88, 66, 78, 60, 82, 51
- Suppose that one person in 100,000 has a particular rare disease for which there is a fairly accurate diagnostic test. This test is correct 99.0% of the time when given to a person selected at random who has the disease; it is correct 99.5% of the time when given to a person selected at random who does not have the disease. Find the probability that a person who
 - tests positive for the disease has actually the disease? [6]
 - tests negative for the disease does not have the disease? [6]
- Let X be the sum of the numbers that appear when a pair of dice is rolled.
 - Find the values of X . [6]
 - Find the expected value of X . [6]
- Solve the recurrence relation with initial conditions $a_0 = a_3 = 2$ and $a_1 = 1 = a_2$. [15]

$$a_n = 7a_{n-1} - 17a_{n-2} + 17a_{n-3} - 6a_{n-4} + 2^n + n$$

6. (a) Are $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$ logically equivalent? [7]
(b) Prove that 3 divides $n^3 + 2n$ whenever n is a non-negative integer. [8]
7. The English alphabet contains 21 consonants and five vowels. How many strings of six lower-case letters contain
- (a) the letter a ? [3]
 - (b) the letters a and b ? [3]
 - (c) the letters a and b in consecutive positions with a preceding b , with all the letters distinct? [3]
 - (d) the letters a and b , where a is somewhere to the left of b in the string, with all the letters distinct? [3]
8. (a) Check the correctness of the argument [7]
A student in the SC612 course has not read the textbook.
Everyone in this course passed the semester exam.
Therefore, someone who passed the semester exam has not read the textbook.
- (b) Check the correctness of the program segment [5]

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x := 2
z := x + y
if y > 0 then
    z := z + 1
else
    z := 0
```

with respect to the initial assertion $y = 3$ and the final assertion $z = 6$.